

TriZOD — Final Pipeline & Re-Referenced Dataset

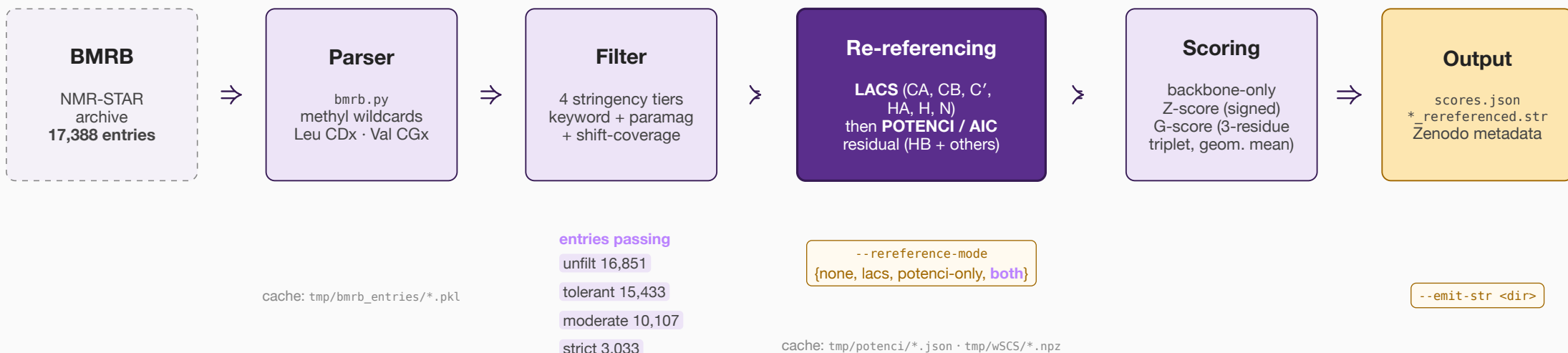
Finalized pipeline · α -synuclein case study · chemical-shift perturbations

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6 May 2026

TUM · TriZOD project meeting

TriZOD pipeline at a glance



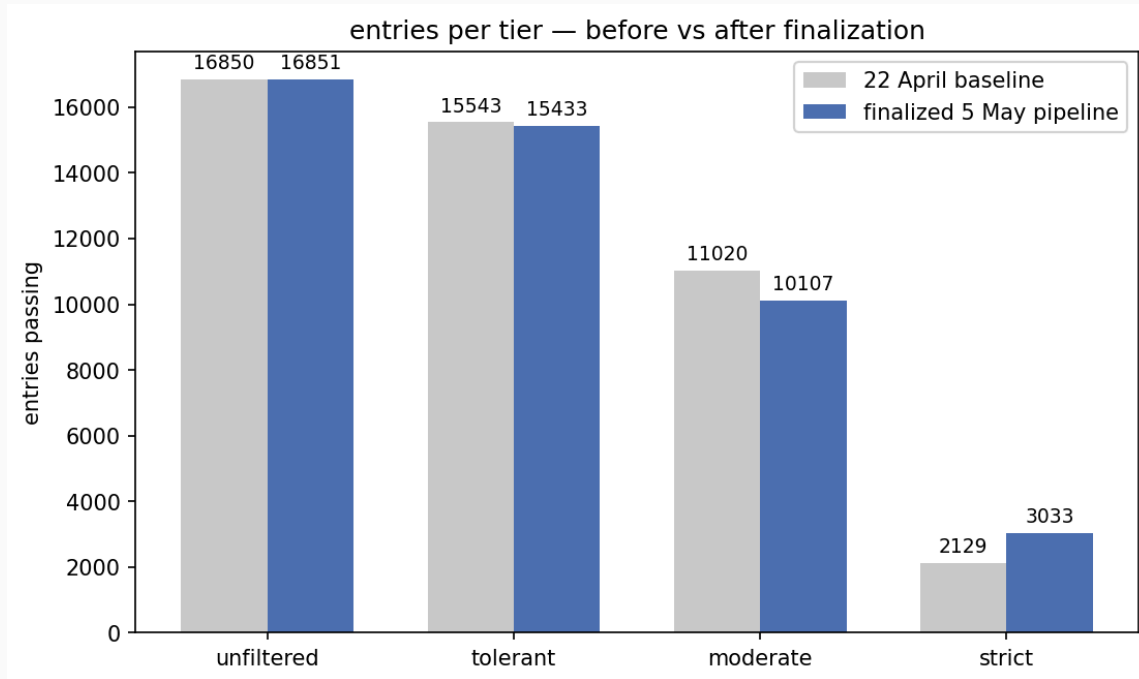
Next steps:

- exclude multi-molecule (bound) entries that distort G-scores
- per-sequence representative pick (best conditions → median G-score)
- mmseqs2 sequence clustering for ML train/val/test split

What's new since 22 April

- Methyl wildcards in the parser (Leu CD1/CD2 $\rightarrow$ CDx, Val CG1/CG2 $\rightarrow$ CGx)
- LACS pre-correction baked into the scoring pipeline
- New flag: `--rereference-mode {none, lacs, potenci-only, both}` (default both)
- New flag: `--emit-str <dir>` writes one re-referenced NMR-STAR per scored entity
- **Zenodo deposit metadata** in repo (DOI on first tagged release)

Filter improvements and per-tier dataset deltas



Removed denaturant false-positives, fixed the solid-state regex, added a paramagnetic-sample filter, relaxed `min-backbone-shift-types 5→4` in strict, and broadened the Celsius heuristic.

Methyl wildcards: what and why

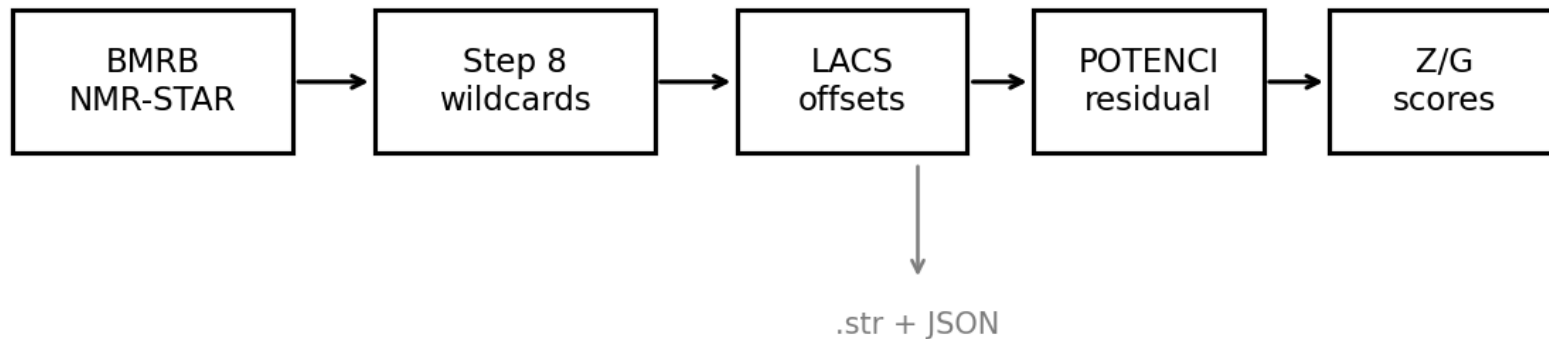
- Leucine: two methyls CD1 + CD2 (γ -carbon).
- Valine: two methyls CG1 + CG2 (β -carbon).
- **Geminal pairs are NMR-equivalent** — most pulse sequences cannot tell them apart.
- BMRB depositors must label one of each pair anyway: arbitrary stereospecificity.

Before: LEU CD1 23.4 ppm · LEU CD2 24.1 ppm (false claim of stereospecificity)

After (ambiguity $\neq$ 1): LEU CDx 23.4 ppm · LEU CDx 24.1 ppm

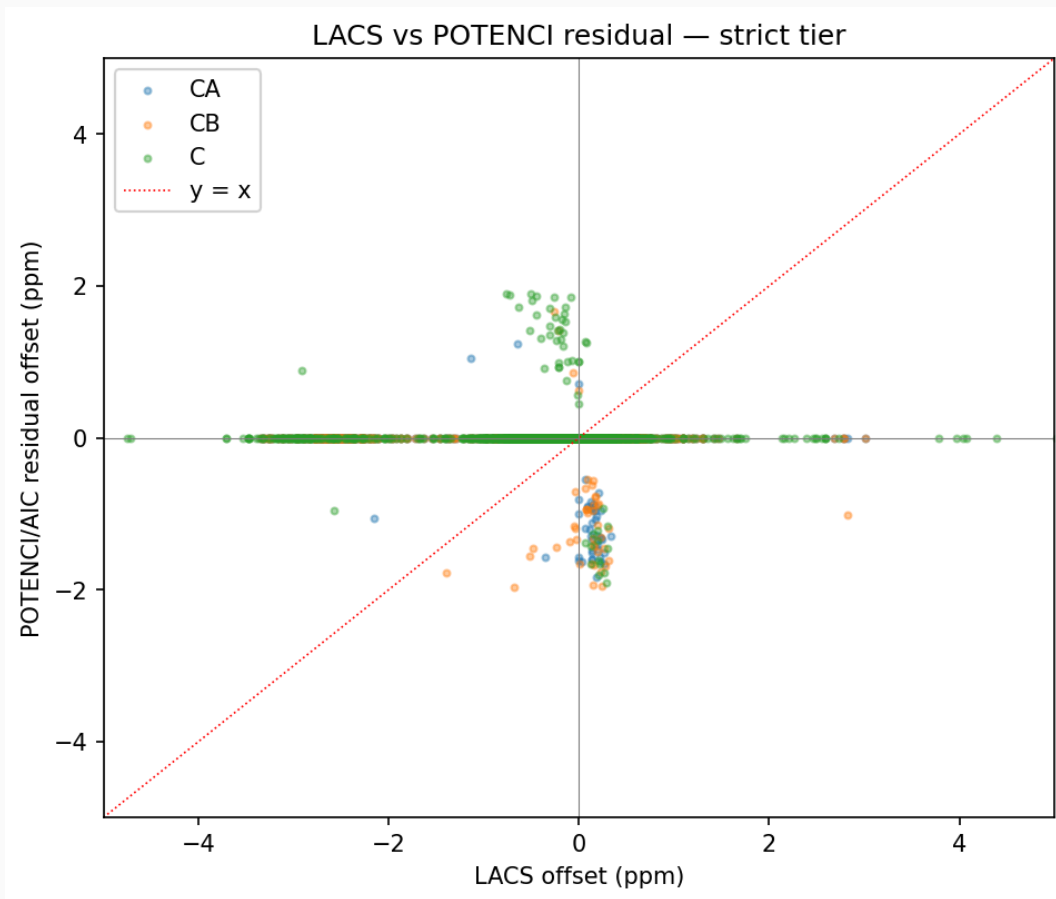
Backbone scoring is unchanged (these are side-chain methyls). The wildcards surface in the emitted `.str` files so downstream auto-assignment tools no longer propagate the false stereospecificity.

Re-referencing in the pipeline



raw shifts → **LACS pre-correction** (Wishart RC tables, robust line fits) → **POTENCI residual** (AIC-gated rolling 9-window) → Z / G-scores. Default mode is both.

LACS vs POTENCI residual capture

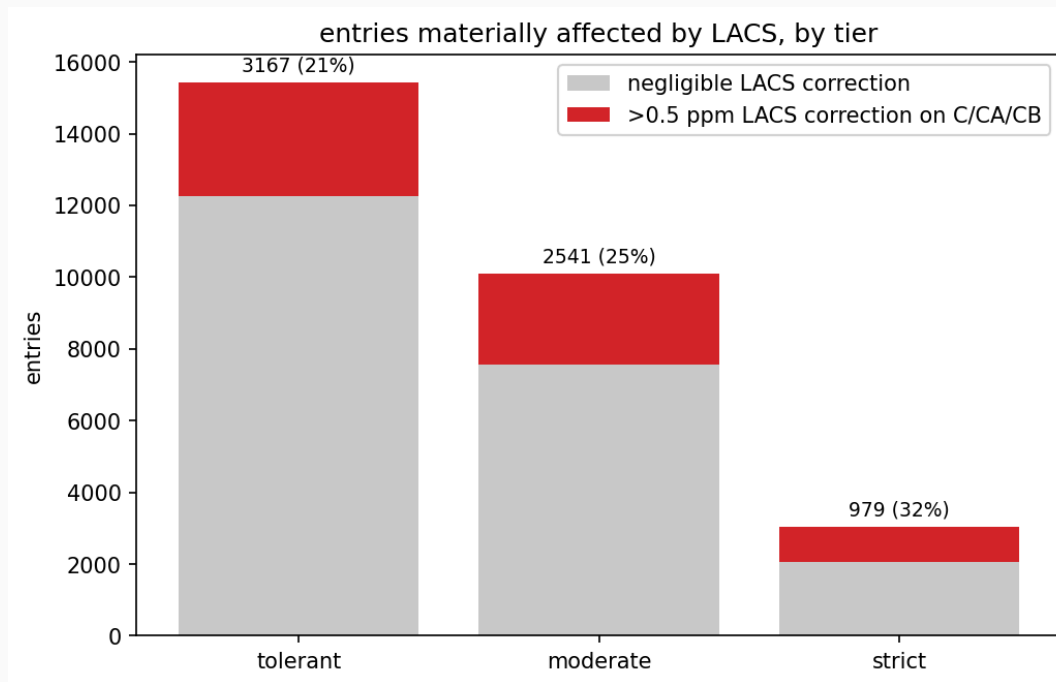


Each dot is **one entry × one atom** (CA / CB / C, strict tier). Not accumulated.

LACS catches the large systematic referencing bias.

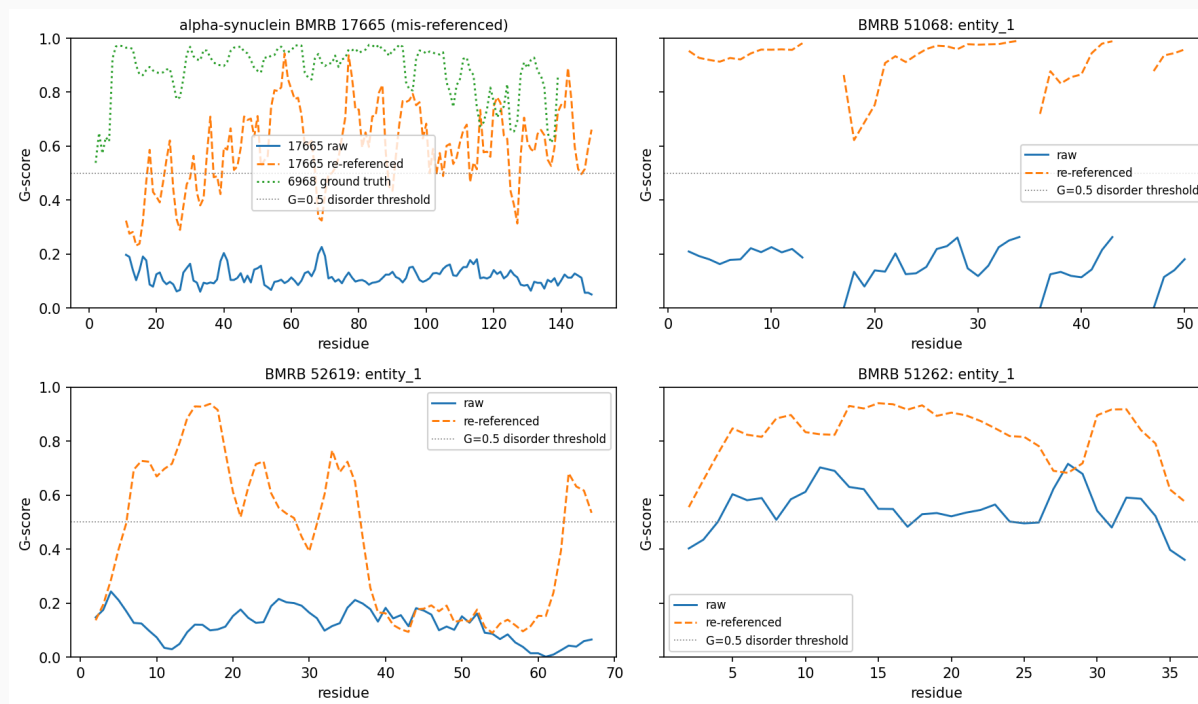
POTENCI/AIC residual mops up the remainder — most points land near the origin once LACS has done its job.

Entries materially affected by LACS



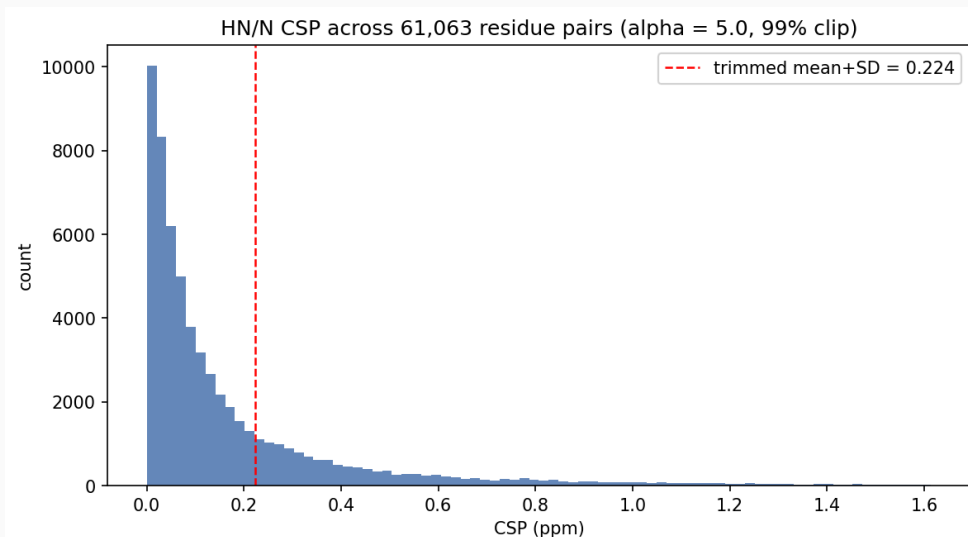
Red = entries with $|\text{LACS offset}| > 0.5$ ppm on at least one of C/CA/CB. Re-referencing meaningfully changes the input shifts for **21%** (tolerant), **25%** (moderate), **32%** (strict).

α -synuclein and top-3 G-score flippers

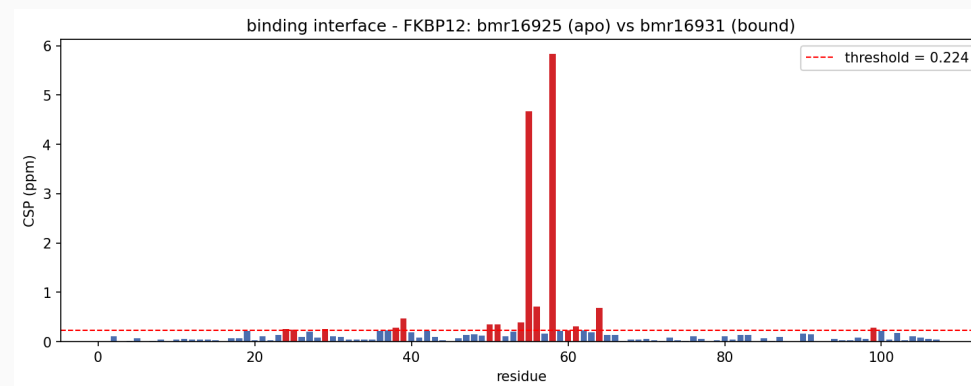


Panel A · BMRB **17665** (α Syn, mis-referenced) raw vs re-referenced + BMRB **6968** (α Syn ground truth). Panels B-D · top-3 flippers BMRB **51068** / **52619** / **51262**. Gaps come from $k = 0$ triplets (insufficient comparable shifts $\rightarrow$ G-score is NaN).

Chemical-shift perturbations (residue-level)

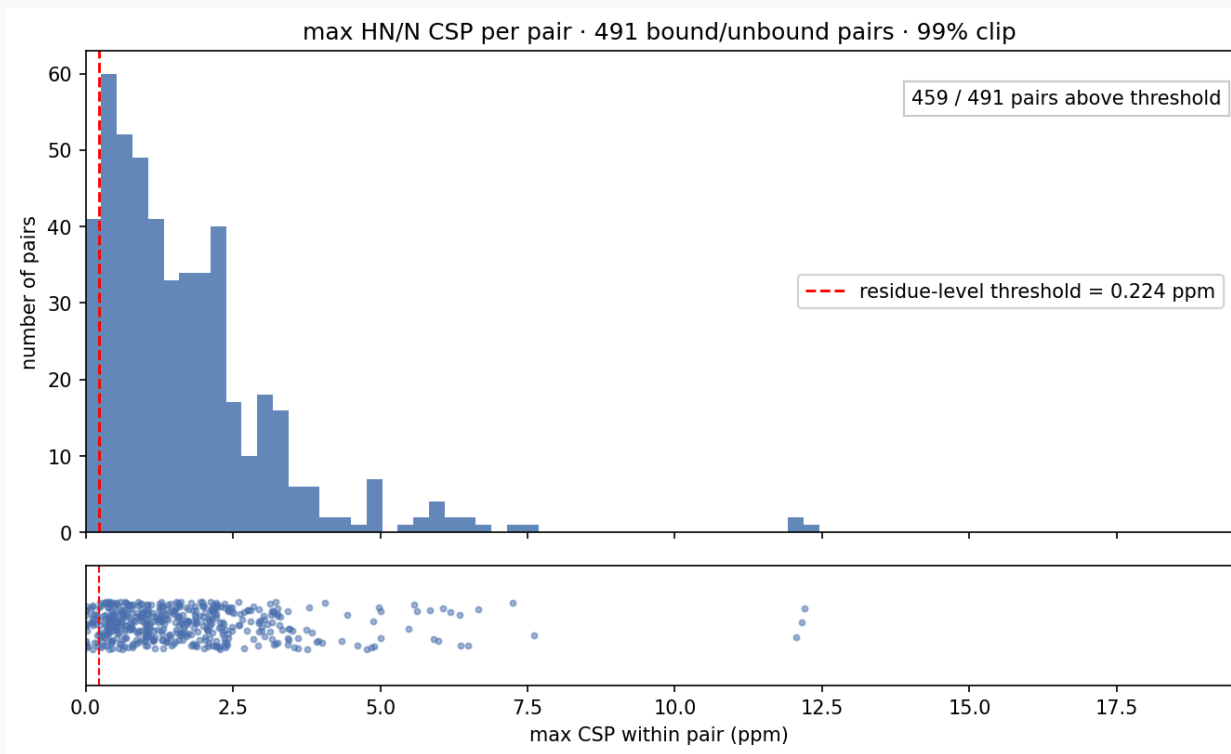


All 61,063 per-residue CSPs across 581 pairs. Threshold **0.224 ppm** = trimmed-mean+SD on the bottom 90% (top 10% dropped so the threshold isn't inflated by interface residues).



FKBP12 apo (bmr16925) vs bound (bmr16931). Two binding-pocket residues (55, 58) far above threshold.

Max CSP per pair · how strong is the strongest binding shift?



One value per pair (581 points): the **maximum** HN/N CSP within each pair. Pairs below the 0.224-ppm threshold are essentially silent; those above are real binding events.

What's still TODO

- **Multi-molecule entries distort G-scores** — exclude entries with a non-polymer or nucleic-acid binding partner from the dataset.
- **Per-sequence representative selection** — for the same protein with multiple BMRB entries, pick the one with best experimental conditions; tiebreak by median G-score.
- **mmseqs2 sequence clustering** for the ML train/val/test split (per the original CheZOD/TriZOD report).
- All three are flagged as TODO in the workflow on slide 2.

Released artefacts

- `data/release/<tier>/scores.json` — per-residue Z/G + LACS + POTENCI offsets
- `data/release/<tier>/str/bmr*_*_rereferenced.str` — re-referenced NMR-STAR
- `.zenodo.json` + `CITATION.cff` — DOI on first tagged release

Per-tier counts after the rerun:

Tier	Entries	Coverage
strict	3,033	17.00%
moderate	10,107	56.64%
tolerant	15,433	86.49%
unfiltered	16,851	94.45%

Next steps

- Push deposit to Zenodo (DOI placeholder until first tag)
- Iterate on Reid's CSP — extend to multi-atom $\Delta\omega_{\text{RMS}}$ (Schumann/Williamson 2008)
- Reid follow-up with John Markley on why BMRB stopped LACS reports (frozen July 2020)
- Possible auto-assignment partner who'd benefit from CDx/CGx outputs